



# National Health Claims Exchange

## *NHCX-PMJAY-HMIS Integration*

# Context: Current Challenges with TMS-Based PMJAY Workflow



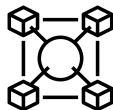
PMJAY claims processing is currently dependent on the TMS 2.0 Provider System, irrespective of hospitals having their own Hospital Management Information Systems (HMIS).

## Challenges



### Duplication of data entry

Parallel data entry in hospital HMIS and TMS increases administrative workload and the risk of errors.



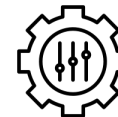
### Lack of System Interoperability

Poor interoperability prevents seamless exchange of beneficiary, clinical, and claims data across systems.



### Single-System Dependency

Exclusive reliance on a single provider system impacts scalability and operational flexibility



### Restricted Innovation and Customisation

System constraints restrict innovation and flexibility for hospitals and technology partners

# Solution: Transition to ABDM-Enabled HMIS Integrated with NHCX



## Elimination of Mandatory TMS Dependency

Unlike the current model where PMJAY processing must be done on TMS 2.0, hospitals can operate PMJAY workflows entirely within their own HMIS without mandatory portal usage. Claims move directly from hospital systems to payer systems via NHCX.

## Single-Source PMJAY Data Capture

PMJAY data is recorded once within the hospital HMIS and reused for claims processing, removing the need to re-enter or reconcile data across systems (thereby reducing errors)

## Greater Choice of PMJAY-Compliant Systems

Hospitals are no longer tied to a single PMJAY processing system and can choose any ABDM-enabled HMIS

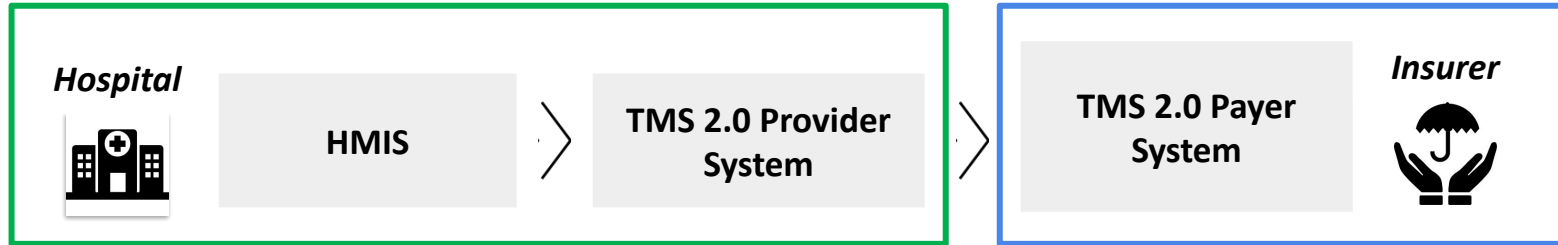
## Scalable PMJAY Operations for High-Volume Hospitals

HMIS-based integrations support higher transaction volumes more reliably than portal-based workflows.

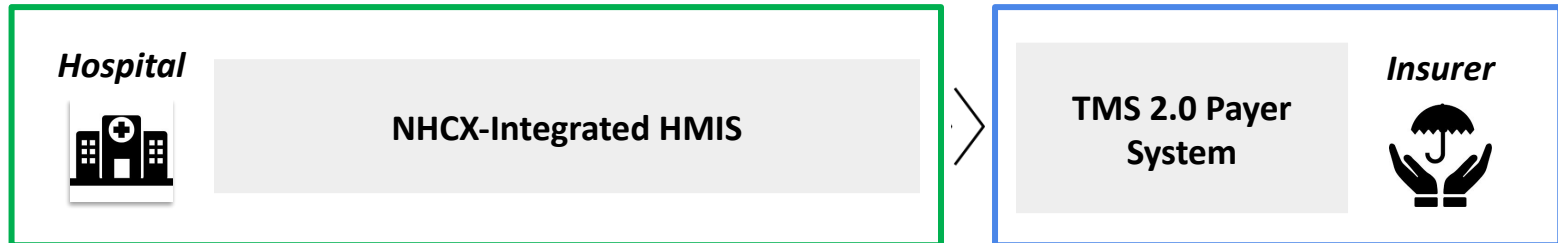
# Solution: Transition to ABDM-Enabled HMIS Integrated with NHCX



**Current scenario:**



**With NHCX Integration:**



# Benefits: What this means for PMJAY-empowered Hospitals



## PMJAY Workflow Embedded in Hospital Operations

PMJAY cases can be handled as part of routine hospital workflows within HMIS, rather than as a separate, portal-driven activity. Hospital PMJAY operations are no longer impacted by portal downtime, access constraints, or concurrent user limitations.

## Availability of Structured Data

Availability of structured data will enable automation of Pre-Auth/Claim resulting in low processing cost and time. This will reduce manual errors and improve the data quality (enabling basic requirement for use of any AI model) as well as aid in fraud control.

## Improved Internal Audit and Financial Management

All PMJAY-related clinical and financial records reside within hospital systems, simplifying audits, internal reviews, and compliance. PMJAY claims data flows natively into hospital billing and accounting systems, improving reconciliation and financial reporting.

## Simplified Training and User Management

Staff require training on a single system instead of managing separate user roles and workflows across HMIS and TMS.

# NHCX-PMJAY-HMIS Integrator Journey (Sandbox)



## 1. Get Compliant for ABDM M1

- ❑ Send Request to access Sandbox APIs
- ❑ Integrate with Sandbox APIs for Milestone 1
- ❑ Functional Testing/ WASA (Security audit)
- ❑ HTC Demo
- ❑ Go Live for ABDM M1

## 2. NHCX Sandbox Registration

- ❑ Register on NHCX Sandbox using ABDM Client ID
- ❑ Generate participant ID
- ❑ Create public and private key
- ❑ Review NHCX Documentation
- ❑ Comprehensively understand the API flows

## 3. NHCX-PMJAY-HMIS integration

- ❑ Test API flows with NHCX dummy payer (recommended)
- ❑ Implement all usecase APIs including biometric authentication and structured data exchange
- ❑ Ensure all cases, including happy flow are tested internally

## 3. Sandbox Exit Testing

- ❑ Internal Demo: Test and Showcase Interoperability, FHIR Bundle Validation
- ❑ PMJAY Team Demo
- ❑ WASA (security audit)
- ❑ HTC Demo
- ❑ Fill NHCX Sandbox Exit Form

**DISCLAIMER: THIS JOURNEY IS TENTATIVE AND NOT FINAL.**

# Additional Requirements for NHCX-PMJAY-HMIS



| Item                                               | Description                                                                                                                                                                                                                                                                                                                                                            |
|----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Utilising the Insurance Plan Response</b>       | <ul style="list-style-type: none"><li>The PMJAY scheme extensively uses the InsurancePlan FHIR bundle, which is configured at the hospital level and serves as a key reference for scheme operations. As this configuration drives several downstream workflows, it is critical for enabling subsequent steps in the pre-authorisation and claims lifecycle.</li></ul> |
| <b>Biometric authentication of the Beneficiary</b> | <ul style="list-style-type: none"><li>The PMJAY scheme mandates biometric authentication of a beneficiary during registration, treatment and discharge. This is an additional API that has to be implemented.</li></ul>                                                                                                                                                |
| <b>Structured data exchange</b>                    | <ul style="list-style-type: none"><li>For a HMIS to process PMJAY claims, it is mandatory to send supporting health information in ABDM-defined structured Health Information Types.</li></ul>                                                                                                                                                                         |
| <b>Query flow</b>                                  | <ul style="list-style-type: none"><li>The PMJAY payer does not use the Communication API for queries. Instead, a query is raised by the payer with the relevant workflow ID for preauth/claim which has to be responded to by the provider with the relevant workflow ID (preauth/claim bundle structure remains the same).</li></ul>                                  |

# Additional Requirements for NHCX-PMJAY-HMIS



| NHCX normal flow           | NHCX-PMJAY-HMIS flow                                                                                                                                                                               |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Get Insurance Plan         | Same                                                                                                                                                                                               |
| Get Policy                 | Same                                                                                                                                                                                               |
| Coverage eligibility check | Same                                                                                                                                                                                               |
| Pre-Auth                   | Same (but before pre-auth need to do mandatory biometric authentication of beneficiary)                                                                                                            |
| Communication Request      | Communication API is not used. Instead, a query is raised by the payer with the relevant workflow ID for preauth/claim which has to be responded to by the provider with the relevant workflow ID. |
| Claim Submission           | Same                                                                                                                                                                                               |
| Payment Notice             | Same                                                                                                                                                                                               |
| Status Check               | Same                                                                                                                                                                                               |

---

# Thank You